

THEORY ASSIGNMENT

Course: Full Stack Web Development
Course Code: 23CM4121
Topic: NPM and package.json
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1. QUESTION

Explain NPM and package.json in Node.js. Describe the steps involved in installing an external package and using it in a Node.js application.

2. ANSWER

What is NPM?

NPM (Node Package Manager) is the default package manager for Node.js, and it is the world's largest software registry. NPM essentially consists of two main parts:

- **The CLI (Command Line Interface):** The tool that developers use to interact with the NPM registry from their terminal (e.g., for installing, updating, or removing packages).
- **The Online Registry:** A massive public database where developers publish and share their open-source JavaScript packages and libraries.

What is package.json?

The *package.json* file is the heart of any Node.js project. It is a JSON (JavaScript Object Notation) file that holds critical metadata about the project and is used to manage its dependencies, scripts, and versioning.

Key roles of *package.json* include:

- **Project Identity:** Contains the project name, version, description, author, and license.
- **Dependency Management:** Lists all the external packages (under *dependencies* and *devDependencies*) required for the project to run. This ensures anyone cloning the repository can install the exact same versions using *npm install*.
- **Scripts:** Allows developers to define custom command-line scripts (like *npm start* or *npm test*) to automate tasks.

Steps to Install and Use an External Package

To demonstrate, we will install and use a popular utility library called **lodash**.

Step 1: Initialize the Project

First, navigate to your project folder in the terminal and run the initialization command. The *-y* flag automatically creates a default *package.json* file without asking interactive questions.

```
npm init -y
```

Step 2: Install the External Package

Use the *npm install* command followed by the package name. This command will download the package into a *node_modules* folder and automatically add it to the *dependencies* object in your *package.json*.

```
npm install lodash
```

Step 3: Require and Use the Package in Code

Create a JavaScript file (e.g., *app.js*) and use the *require()* function to import the module. Once imported, you can use its methods.

```
// app.js
const _ = require('lodash'); // Importing the installed 'lodash' package

const numbers = [10, 5, 100, 2, 1000];

// Using a lodash utility method to find the maximum number
const maxNum = _.max(numbers);

console.log(`The maximum number is: ${maxNum}`); // Output: The maximum number is: 1000
```

Step 4: Execute the Application

Run the file using the Node.js runtime from the terminal.

```
node app.js
```